Advanced Programming Practice · 21CSC203P

Unit 1 · Session 6 · SLO 2 — Solved Worksheet

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Unit – 1 : Session – 6 : SLO – 2 | Modular Programming

Activity (Solutions)

• 1) Program broken into reusable methods: static int readN(){ return new java.util.Scanner(System.in).nextInt(); } static boolean isEven(int n){ return n%2==0; } static void report(int n){ System.out.println(n + (isEven(n) ? " even" : " odd")); }

• 2) Main + two helpers: public class Main { static int square(int x){ return x*x; } static void printResult(int v){ System.out.println("Result: " + v); } public static void main(String[] a){ printResult(square(6)); } } Output: Result: 36.

• 3) Code reuse via static methods: static methods belong to the class, not instances, so any code can call ClassName.method() without creating objects — Java's Math.sqrt(), Math.pow(), Integer.parseInt() are examples; our own utility classes (e.g., a Utils class with static helpers) give the same reuse across the program.